

INTERNATIONAL SCHOOL

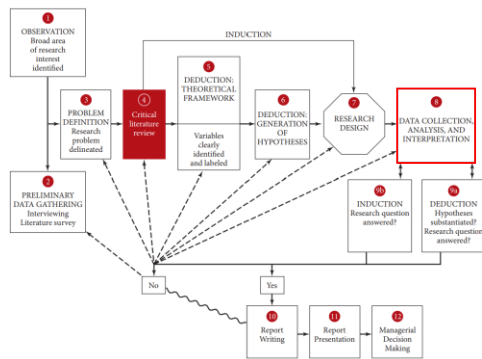
Data Collection Methods

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RESEARCH WORKFLOW



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LEARNING OBJECTIVES

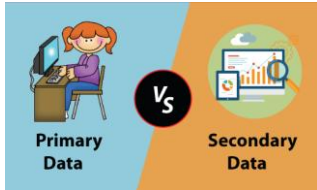
- 1) Secondary data collection methods
- 2) Primary data collection methods
- 3) Choose the data collection method for your topic

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SOURCES OF DATA

- Data be obtained from primary or secondary sources:
 - ✓ Primary data refers to information obtained firsthand by the researcher on the variables of interest for the specific purpose of the study
 - ✓ Secondary data refers to information gathered from sources already existing



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SOURCES OF DATA

Points	Primary Data	Secondary Data
Meaning	Primary data is collected directly from first-hand experience for a specific research project.	Secondary data is pre-collected by someone else for a different purpose.
Main data sources	Interviews, questionnaires, observations, experiments, case studies.	Previous research, internet, government reports, official statistics, web information, etc.
Data Time	Real-time data.	Past data.
Specific to the Researcher Needs	Always specific to the researcher's needs.	Often not specific to the researcher's needs.

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SOURCES OF DATA

Points	Primary Data	Secondary Data
Costs	Expensive.	Low cost or free.
Level of Control Over Data Quality	Higher level of control.	Lack of control over data quality.
Time Consuming	More time-consuming.	Less time-consuming.
Proprietary Information	You own the data, keeping it hidden from competitors.	Competitors may also have access to the data.

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SECONDARY SOURCES OF DATA



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SECONDARY SOURCES OF DATA

Example: Relationships between inflation and household consumption expenditure in Vietnam in 2016-2026.

Variable Type	Specific Variable Name	Data Source (Secondary Data)
Dependent Variable	Total Retail Sales of Goods and Services	General Statistics Office (GSO) - "Trade and Services" section in Monthly Socio-economic Reports.
Independent Variable	Consumer Price Index (CPI)	General Statistics Office (GSO) - "Price Index" section or Annual Statistical Yearbooks.
Independent Variable	GNI/GDP per Capita (Income)	GSO - Vietnam Household Living Standard Survey (VHLSS) or World Bank Open Data.
Control Variable	VND/USD Exchange Rate	State Bank of Vietnam (SBV) or historical data from Vietstock / Investing.com.

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ETHICS IN SECONDARY DATA COLLECTION

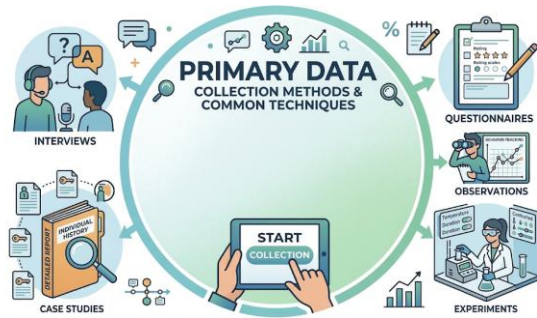
1. If the data is freely available, the ownership of the original data must be acknowledged.
2. If the data is not freely available, written permission for the use of the data must be obtained.

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PRIMARY DATA COLLECTION METHODS



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INTERVIEWS

- A widely used method of collecting data in research is **to interview respondents TO OBTAIN INFORMATION on an issue of interest**
- An interview is a guided, purposeful conversation between two or more people

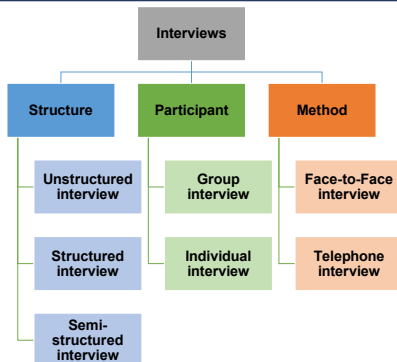


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INTERVIEWS



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Unstructured interviews

- The interviewer does **not** enter the interview setting with a **planned sequence of questions** to be asked of the respondent
- A possible objective of an unstructured interview is to bring some **preliminary issues** to the surface
 - ✓ So that the researcher can determine what factors need further in-depth investigation



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Unstructured interviews

- What do you like about working here?
- If you were to tell me which aspects of your job you like and which you do not, what would they be?
- Tell me something about the reward systems in this place.
- If you were offered a similar job elsewhere, how willing would you be to take it and why?
- & so on

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Structured interviews

- Structured interviews are those conducted when it is known at the **outset what information is needed**
- The content can be prepared in advance, consists of:
 - ✓ **An introduction:** the interviewer introduces him or herself, the purpose of the interview, permission to record the interview, assures confidentiality, etc...
 - ✓ **A set of topics (usually questions) in a logical order:** first "warm-up" questions (which are easy to answer and non-threatening) and then the main questions covering the purpose of the interview, etc...
 - ✓ **Suggestions for probing questions:** follow-up questions that are used when the first answer is unclear or incomplete, the interviewer does not fully understand the answer, etc...

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Semi- structured interviews

- A semi-structured interview has a **set of predetermined questions** or topics to be covered, but there is also **flexibility to explore** new ideas and probe deeper into certain responses.
- Semi-structured interviews allow for a more conversational and flexible interaction between the interviewer and the interviewee.

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Interviews

	Structured interview	Semi-structured interview	Unstructured interview
Fixed questions	✓	✓	✗
Fixed order of questions	✓	✗	✗
Fixed number of questions	✓	✗	✗
Option to ask additional questions	✗	✓	✓

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Group interviews

- Interviews may be conducted on a **group basis**:

✓The interviewer puts open questions to a group of participants



- Focus groups **consist of eight to ten members** with a moderator leading the discussions on a particular topic, concept, or product

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OBSERVATION

- **Researchers be interested in:**

- ✓ The way workers carry out their jobs;
- ✓ The impact of new manufacturing techniques on employee activity;
- ✓ How consumers watch commercials, use products, ...



=> **OBSERVATION: is a useful and natural technique to collect data on actions & behavior**

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Definition and purpose of observation

- **Observation concerns the planned watching, recording, analysis, and interpretation of behavior, actions, or events**

- **These may be distinguished by four key dimensions:**

- ✓ (1) **Control** (are the observations conducted in an artificial or natural setting?)
- ✓ (2) **Whether the observer is a member of the group that is observed or not** (participant versus nonparticipant observation)
- ✓ (3) **Structure** (to what extent the observation is focused, predetermined, systematic, and quantitative in nature)
- ✓ (4) **Concealment** (whether the participants are told that they are being observed)

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Controlled - Uncontrolled observations

- **A distinction can be made between observation conducted in controlled (or artificial) and uncontrolled (or natural) settings**

- **Controlled observation (or artificial) occurs when observational research is carried out under carefully arranged conditions**

- **Uncontrolled observation (or natural) is an observational technique that makes no attempt to control, manipulate, or influence the situation**

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Participant - Nonparticipant observations

- **The researcher can play one of two roles while gathering observational data – that of a nonparticipant or a participant observer**
- **Nonparticipant observation:** The researcher is never directly involved in the actions of the actors, but observes them from outside the actors' visual horizon
 - ✓ Example: via a mirror or a camera

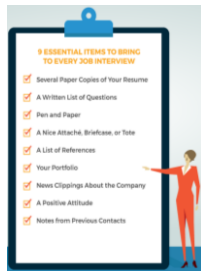


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Structured - Unstructured observations

- **Structured observational study:** the observer has a predetermined set of categories of activities or phenomena planned to be studied
- **Unstructured observational study:** observer will record practically everything that is observed



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Advantages and disadvantages of observation

- **ADVANTAGES:**
 - ✓ The data are generally reliable and free from respondent
 - ✓ Directness
 - ✓ Observation allows the researcher to gather behavioral data without asking questions
- **DISADVANTAGES:**
 - ✓ The observer to be physically present
 - ✓ Slow, tedious, expensive
 - ✓ Long periods => might bias the recorded data
 - ✓ Cognitive processes of individuals can't be captured
 - ✓ Observers have to be trained in what and how to observe, and ways to avoid observer bias

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QUESTIONNAIRES

- **Questionnaire is a written set of questions to which respondents record their answers, usually within rather closely defined alternatives**
- **Questionnaires are generally designed to collect large numbers of QUANTITATIVE DATA**
- **Types of questionnaires:**
 - ✓(1) Personally administered questionnaires
 - ✓(2) Mail questionnaires
 - ✓(3) Electronic / Online questionnaires



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Personal questionnaires

- **ADVANTAGES:**
 - ✓Researcher can collect all the completed responses within a short period of time
 - ✓Less expensive and consumes less time than interviewing
- **DISADVANTAGES:**
 - ✓The researcher may introduce a bias by explaining questions differently to different people
 - ✓Participants may be in fact answering different questions



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Mail questionnaires

- **Main advantage:**
 - ✓Wide geographical area can be covered in survey
 - ✓Respondents can complete at their convenience: homes, work-places
- **Disadvantages:**
 - ✓The return rates are typically low: ~30%
 - ✓Respondent's Doubts: difficult to be clarified



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Electronic & Online questionnaires

- **Main advantage: Easy and Fast**
 - ✓ All you have to do is to email the invitations, post a link on a website or personal blog, or use social networks
- **Task requiring:** Web authoring programs, HTML codes, scripting programs
- **Google form:** Drive => New => Google form

Google Form Sample

Here are all of the question types

This is a text question
Limited space for typed responses. Great space for names or short answer.

Your answer

This is a paragraph question
Extended space for longer answers.

Your answer

This is a multiple choice question
May allow one of the choices or request to 'other'.

☐ Choice 1

☐ Choice 2

☐ Other

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Guidelines for Questionnaire design

- 1) **Principles of wording: refer to factors as:**
 - ✓ **Content and purpose of the questions:** subjective feelings or objective facts
 - ✓ **Language and wording of the questionnaire**
 - ✓ **Type and form of questions asked:** open or closed question
 - ✓ **Sequencing of the questions:** easy to difficult, general to detail
 - ✓ **Personal information from the respondents:** age, educational level, marital status, and income

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Guidelines for Questionnaire design

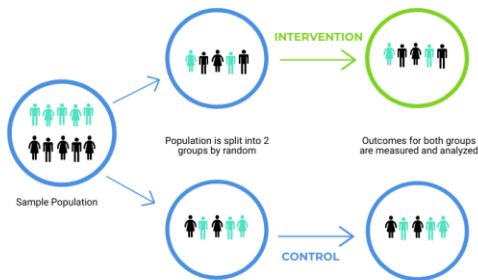
- 2) **Principles of measurement:**
 - ✓ General appearance: Good introduction, Organizing questions, giving instructions and guidance, Good alignment ...
- 3) **Review of questionnaire design:** minimize response bias and measurement errors
- 4) **Pretesting of structured questions:**
 - ✓ Pretesting involves the use of a small number of respondents to test the appropriateness of the questions and their comprehension

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Experiments

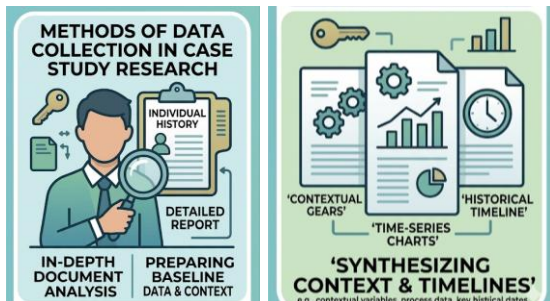


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Case study



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EXERCISE 1

Research topic name: Consumer purchase intentions for organic food in Hanoi and Ho Chi Minh City in 2026.

- **Option A:** Sending **an online survey** (Google Forms) to community groups and apartment resident pages on Facebook.
- **Option B:** **Standing at organic food aisles in supermarkets** to observe selection behavior and conduct quick (5-minute) interviews with customers immediately after they pick a product.
- **Option C:** Gathering and analyzing **sales reports** of organic food from major supermarket chains over the past three years.

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EXERCISE 2

Research topic name: Relationship between the smartphone usage and academic performance in VNUIS in 2026

Choose the research design, data collection method, variables for this research.

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EXERCISE 3

Please write down on paper the name of 01 research topic.

Then identify **which type of data collection method** is suitable for your research topic.

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